

# FBISE SSC-I COMPUTER SCIENCE

## New Model Paper - Complete Detailed Solution

National Curriculum 2022-2023  
(Inclusive Scheme of Studies 2024)

Section A: 60 MCQs with correct answer, reasoning and explanation of every wrong option

Section B: All 6 questions solved, including every OR option

Section C: Both 6-mark questions solved, including both OR options

**New Pattern: 50% MCQs + 50% Descriptive**

**Prepared by Fakhar STEM Sphere**

*Student-friendly, exam-oriented and concept-based solution*

## How to Use This Solution

This solution is designed so that a Class 9 student can understand not only what the correct answer is, but also why it is correct. For every MCQ, the incorrect choices are discussed because scenario-based FBISE questions often test understanding rather than memorization.

**Paper Structure:** Section A = 60 MCQs (30 marks), Section B = 6 short questions x 3 marks (18 marks), and Section C = 2 long questions x 6 marks (12 marks).

For descriptive questions, both the main question and the OR question have been attempted. Students should learn the concepts and reproduce the answer in clear points rather than memorizing wording blindly.

### Quick Contents

- Section A - MCQs 1-40
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- Exam Preparation Checklist

## SECTION A - 60 MCQs WITH COMPLETE REASONING

Each MCQ carries 0.5 mark. The key exam skill is to identify the concept hidden inside the wording or scenario. The explanations below therefore focus on both the correct reasoning and the reason each distractor is unsuitable.

### 1. A computer model used in a weather forecasting system is primarily designed for:

|                                                                  |                                                             |
|------------------------------------------------------------------|-------------------------------------------------------------|
| A. To archive weather data permanently                           | B. To control weather monitoring equipment                  |
| C. To simulate atmospheric conditions and predict future weather | D. To calculate electricity consumption in weather stations |

**Correct Answer: C - To simulate atmospheric conditions and predict future weather**

**Why it is correct:** A weather model represents atmospheric processes mathematically and uses current observations to simulate how temperature, pressure, humidity, wind and other conditions may change. Its main purpose is therefore prediction, not storage or equipment control.

**Why the other options are wrong:**

- A is wrong because archiving is a data-storage function, not the primary purpose of a forecasting model.
- B is wrong because sensors and control systems manage monitoring equipment; the forecasting model interprets data.
- D is wrong because electricity consumption at weather stations is unrelated to forecasting atmospheric conditions.

### 2. What is the primary purpose of HTML in web development?

|                                                    |                                                           |
|----------------------------------------------------|-----------------------------------------------------------|
| A. To design the layout and structure of a webpage | B. To add dynamic behavior and interactivity to a webpage |
| C. To manage databases on a website                | D. To control computer operations                         |

**Correct Answer: A - To design the layout and structure of a webpage**

**Why it is correct:** HTML provides the basic structure and content of a webpage by defining headings, paragraphs, images, links, tables and other elements. In beginner-level web development, it is the structural language of the page.

**Why the other options are wrong:**

- B is mainly the role of JavaScript.
- C is handled by database systems and server-side technologies, not HTML.
- D is the job of the operating system and system software, not a webpage markup language.

### 3. Which tag is used to create a hyperlink in HTML?

|          |          |
|----------|----------|
| A. <img> | B. <a>   |
| C. <p>   | D. <div> |

**Correct Answer: B - <a>**

**Why it is correct:** The anchor tag <a> is used to create a hyperlink. Its href attribute specifies the destination, for example: <a href="page.html">Open Page</a>.

**Why the other options are wrong:**

- A (<img>) displays an image.
- C (<p>) creates a paragraph.
- D (<div>) creates a general container or division.

### 4. Which symbol is commonly used to end statements in JavaScript?

|                  |               |
|------------------|---------------|
| A. Colon (:)     | B. Period (.) |
| C. Semicolon (;) | D. Comma (,)  |

**Correct Answer: C - Semicolon (;)**

**Why it is correct:** A semicolon is commonly written at the end of a JavaScript statement, for example: let x = 10;. JavaScript can sometimes insert semicolons automatically, but semicolon termination is standard and is the expected answer here.

**Why the other options are wrong:**

- A colon is used in other syntactic contexts, not as the normal statement terminator.
- A period is commonly used for property or method access, such as object.name.
- A comma separates items or expressions; it is not the standard statement terminator.

### 5. Why is visualization important in data analysis?

|                                          |                                 |
|------------------------------------------|---------------------------------|
| A. It increases file size                | B. It makes data harder to read |
| C. It helps in understanding data easily | D. It removes unnecessary data  |

**Correct Answer: C - It helps in understanding data easily**

**Why it is correct:** Graphs, charts and diagrams make patterns, comparisons and trends easier to see than long lists of raw values. This helps people interpret data quickly and make decisions.

**Why the other options are wrong:**

- A may happen in some cases but is not the purpose of visualization.
- B is the opposite of the goal of visualization.
- D describes data cleaning, not data visualization.

**6. A student organizes exam marks in a table to easily compare results. What does this represent?**

|                    |                  |
|--------------------|------------------|
| A. Data collection | B. Data analysis |
| C. Data encryption | D. Data backup   |

**Correct Answer: B - Data analysis**

**Why it is correct:** Organizing marks in a table so they can be compared and interpreted is part of data analysis. The student is using the collected values to understand performance.

**Why the other options are wrong:**

- A would mean obtaining the marks in the first place.
- C means converting data into a protected coded form.
- D means creating an additional copy for recovery.

**7. Which programming language is commonly used for web development?**

|               |           |
|---------------|-----------|
| A. HTML       | B. CSS    |
| C. JavaScript | D. Python |

**Correct Answer: C - JavaScript**

**Why it is correct:** JavaScript is a programming language widely used to add logic, interactivity and dynamic behavior to webpages. HTML and CSS are important web technologies, but they are not programming languages in this context.

**Why the other options are wrong:**

- A is a markup language used to structure webpages.
- B is a stylesheet language used to style webpages.
- D is a programming language and can be used on the web, especially on servers, but JavaScript is the expected browser/web-development answer among these choices.

**8. A CSS rule is written as: `body { background-color: yellow; }` What will happen?**

|                        |                                   |
|------------------------|-----------------------------------|
| A. Text becomes yellow | B. Page background becomes yellow |
| C. Image turns yellow  | D. Button color changes to yellow |

**Correct Answer: B - Page background becomes yellow**

**Why it is correct:** The selector `body` targets the page body, and the CSS property `background-color` changes its background. Therefore the visible page background becomes yellow.

**Why the other options are wrong:**

- A would require the color property, such as `color: yellow;`
- C would require styling or filtering the image itself.
- D would require selecting the button and changing its background or color property.

**9. Students need to organize and show relationships between AI topics such as machine learning, computer vision and natural language processing. Which tool is most suitable?**

|              |                 |
|--------------|-----------------|
| A. Flowchart | B. Trace tables |
| C. Algorithm | D. Concept maps |

**Correct Answer: D - Concept maps**

**Why it is correct:** A concept map is designed to show concepts and the relationships among them, so it is ideal for organizing branches of Artificial Intelligence in a connected visual form.

**Why the other options are wrong:**

- A flowchart shows the sequence of steps and decisions in a process.
- A trace table follows variable values while an algorithm runs.
- An algorithm is a step-by-step solution procedure, not a relationship map.

**10. Which scenario best illustrates the transition from Process to Output in an IPO chart for a weather app?**

|                                               |                                                             |
|-----------------------------------------------|-------------------------------------------------------------|
| A. Entering the city name and clicking search | B. Calculating the average weekly temperature from raw data |
|-----------------------------------------------|-------------------------------------------------------------|

|                                                            |                                                       |
|------------------------------------------------------------|-------------------------------------------------------|
| C. Converting calculated data into a visual 7-day forecast | D. Refreshing the app to check for new server updates |
|------------------------------------------------------------|-------------------------------------------------------|

**Correct Answer: C - Converting calculated data into a visual 7-day forecast**

**Why it is correct:** In an IPO model, raw data is input, calculations are processing, and the displayed result is output. Turning calculated weather values into a visible seven-day forecast is therefore the transition from processing to output.

**Why the other options are wrong:**

- A is mainly an input/user-action step.
- B describes processing only.
- D is a maintenance/update action and is not the normal output stage of the weather calculation.

**11. In a voice assistant system, Natural Language Processing (NLP) is used to:**

|                                                  |                                                        |
|--------------------------------------------------|--------------------------------------------------------|
| A. Convert audio signals into electrical power   | B. Extract meaning and context from human speech input |
| C. Encrypt user voice data for security purposes | D. Connect the assistant to external hardware devices  |

**Correct Answer: B - Extract meaning and context from human speech input**

**Why it is correct:** NLP helps a computer work with human language. In a voice assistant, once spoken words are recognized, NLP helps interpret their meaning, intent and context so the system can respond appropriately.

**Why the other options are wrong:**

- A is an electrical/hardware conversion task.
- C is a cybersecurity function.
- D concerns hardware interfacing and is not the purpose of NLP.

**12. Which technology is used to build systems like chatbots that can interact with users and provide automated responses?**

|                       |                    |
|-----------------------|--------------------|
| A. Machine learning   | B. Expert system   |
| C. Digital assistance | D. Computer vision |

**Correct Answer: C - Digital assistance**

**Why it is correct:** The scenario describes a digital assistant style system that communicates with users and gives automated responses. Chatbots and virtual assistants are common forms of digital assistance.

**Why the other options are wrong:**

- A can power advanced chatbots internally, but it is a learning technique rather than the user-facing type of system described.
- B focuses on expert rule-based decision making in a specific domain rather than general conversational assistance.
- D is used to interpret images and video, not primarily text conversation.

**13. A student tests a solution by executing it and fixing any errors that appear. Which stage of the problem-solving process is being carried out?**

|                        |                |
|------------------------|----------------|
| A. Decomposition       | B. Debugging   |
| C. Pattern recognition | D. Abstraction |

**Correct Answer: B - Debugging**

**Why it is correct:** Debugging means finding, identifying and correcting errors in a program or solution after testing or execution.

**Why the other options are wrong:**

- A means breaking a large problem into smaller parts.
- C means finding repeated similarities or trends.
- D means focusing on essential information and ignoring unnecessary detail.

**14. A student cannot identify an error in a program. The teacher advises reviewing the flowchart created before coding. How can the flowchart help?**

|                                                             |                                                  |
|-------------------------------------------------------------|--------------------------------------------------|
| A. It will automatically correct the error                  | B. It will execute the program                   |
| C. It will show the logical sequence of steps and decisions | D. It will convert the program into machine code |

**Correct Answer: C - It will show the logical sequence of steps and decisions**

**Why it is correct:** A flowchart gives a visual representation of the program logic. By following the sequence and decisions, a student can compare the intended logic with the code and locate where the reasoning may have gone wrong.

**Why the other options are wrong:**

- A flowchart does not repair program code automatically.
- A flowchart is not an execution environment.

- Conversion to machine code is performed by language-processing software, not a flowchart.

**15. A student creates a webpage and wants to display a notification popup. Which JavaScript function will achieve this?**

|            |              |
|------------|--------------|
| A. alert() | B. print()   |
| C. show()  | D. message() |

**Correct Answer: A - alert()**

**Why it is correct:** The JavaScript alert() function displays a simple browser message box, for example: alert("Welcome!");.

**Why the other options are wrong:**

- print() is associated with printing, not a basic notification popup.
- show() is not the standard built-in JavaScript function for this purpose.
- message() is not the standard built-in JavaScript popup function.

**16. A student is using several software applications at the same time and the computer becomes slow. Which component should be upgraded to improve performance?**

|             |          |
|-------------|----------|
| A. ROM      | B. RAM   |
| C. Register | D. Cache |

**Correct Answer: B - RAM**

**Why it is correct:** Running several applications simultaneously requires working memory. Increasing RAM gives the system more space to keep active programs and their data available, reducing slowdowns caused by insufficient memory.

**Why the other options are wrong:**

- ROM stores firmware and is not used as normal working memory.
- Registers are tiny CPU storage locations and are not a practical system upgrade for multitasking.
- Cache is very fast but small; the expected general upgrade for many simultaneous applications is more RAM.

**17. A student studies previous question papers to identify similarities in question types and predict possible future questions. Which computational-thinking concept is being used?**

|                  |                        |
|------------------|------------------------|
| A. Decomposition | B. Pattern recognition |
| C. Abstraction   | D. Algorithm           |

**Correct Answer: B - Pattern recognition**

**Why it is correct:** Pattern recognition means finding repeated structures, similarities or trends. Looking across past papers for recurring question types is a direct example.

**Why the other options are wrong:**

- A would involve breaking the task into smaller parts.
- C would involve ignoring unimportant details and focusing on essential features.
- D is a step-by-step procedure for solving a problem.

**18. Which of the following classifications represents categorical data?**

|                 |               |
|-----------------|---------------|
| A. Quantitative | B. Binary     |
| C. Qualitative  | D. Continuous |

**Correct Answer: C - Qualitative**

**Why it is correct:** Categorical data describes qualities, names or groups rather than measured numerical quantities. This is commonly called qualitative data, for example eye color or type of vehicle.

**Why the other options are wrong:**

- A refers to numerical data.
- B simply means two possible states and can be represented in different ways; it is not the general classification asked here.
- D is numerical data that can take values over a continuous range.

**19. Big data helps organizations to:**

|                     |                          |
|---------------------|--------------------------|
| A. Ignore customers | B. Reduce information    |
| C. Avoid technology | D. Make better decisions |

**Correct Answer: D - Make better decisions**

**Why it is correct:** By analyzing very large datasets, organizations can identify trends, customer behavior, risks and opportunities. These insights support more informed decisions.

**Why the other options are wrong:**

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- A is contrary to customer analysis.
- B is not the goal; useful information is extracted from large data.
- C is the opposite of how big-data systems operate.

**20. A young entrepreneur evaluates a business idea by gathering customer feedback and studying competitors before making a financial investment. Which stage of the entrepreneurial process is being applied?**

|               |              |
|---------------|--------------|
| A. Production | B. Marketing |
| C. Empathize  | D. Budgeting |

**Correct Answer: C - Empathize**

**Why it is correct:** The entrepreneur is trying to understand users, their needs and the surrounding market before building or investing. In design-thinking language, this matches the empathize stage.

**Why the other options are wrong:**

- A refers to making or delivering the product.
- B is the promotion and selling stage.
- D concerns financial planning rather than understanding users and their needs.

**21. A developer wants to style a webpage by changing colors, fonts and layout. Which language is used for this purpose?**

|        |               |
|--------|---------------|
| A. CSS | B. HTML       |
| C. ASP | D. JavaScript |

**Correct Answer: A - CSS**

**Why it is correct:** CSS (Cascading Style Sheets) controls the visual presentation of webpages, including colors, fonts, spacing, positioning and layout.

**Why the other options are wrong:**

- B provides webpage structure and content.
- C is associated with server-side web technology, not the standard styling language.
- D is mainly used for behavior and interactivity.

**22. Which technology is used in mobile phones for face recognition?**

|                    |                     |
|--------------------|---------------------|
| A. Computer Vision | B. NLP              |
| C. Robotics        | D. Machine learning |

**Correct Answer: A - Computer Vision**

**Why it is correct:** Face recognition requires a computer to analyze and interpret visual information from a camera. This is a direct application of computer vision.

**Why the other options are wrong:**

- B processes human language.
- C deals with physical robots and automated machines.
- D may support recognition internally, but the field directly concerned with understanding faces in images is computer vision.

**23. Cloud computing primarily offers computing resources and services over the internet. What does it mainly provide?**

|                                               |                                     |
|-----------------------------------------------|-------------------------------------|
| A. Online storage and computing services      | B. To replace all computer hardware |
| C. To provide internet browsing services only | D. Only data backup solutions       |

**Correct Answer: A - Online storage and computing services**

**Why it is correct:** Cloud computing gives users internet-based access to storage, servers, software, databases and computing power. It can scale resources without the user owning all infrastructure locally.

**Why the other options are wrong:**

- B is too absolute; local devices and hardware are still needed.
- C is far too limited; cloud services are not just web browsing.
- D is also too limited; backup is only one possible cloud service.

**24. A school installs the same licensed copy of Microsoft Windows on several computers using one activation key without buying separate licenses for each system. What type of software usage does this represent?**

|                     |                         |
|---------------------|-------------------------|
| A. Freeware         | B. Shareware            |
| C. Pirated Software | D. Open Source software |

**Correct Answer: C - Pirated Software**

**Why it is correct:** Using paid software beyond the permissions of its license without purchasing the required licenses is unauthorized copying/use and is treated as software piracy in this context.

**Why the other options are wrong:**

- A is software legally distributed free of charge.
- B is software distributed for trial or limited use under its license conditions.
- D is software distributed under an open-source license with source code access rights.

**25. A student receives an email that appears to be from a bank and urgently asks for a password and personal information through a link. What is this type of email known as?**

|           |             |
|-----------|-------------|
| A. Spam   | B. Phishing |
| C. Cookie | D. Spyware  |

**Correct Answer: B - Phishing**

**Why it is correct:** Phishing messages impersonate trusted organizations and try to trick victims into revealing passwords, banking details or other sensitive information through deceptive links or forms.

**Why the other options are wrong:**

- A is unsolicited bulk messaging and may not attempt credential theft.
- C is data stored by a website in the browser; it is not an email attack.
- D is malicious software that secretly monitors a device; it is not the name of the deceptive email itself.

**26. How does the Control Unit manage instructions while a program is being executed?**

|                                                               |                                                                   |
|---------------------------------------------------------------|-------------------------------------------------------------------|
| A. It performs mathematical calculations and logic operations | B. It directs and coordinates all CPU activities during execution |
| C. It permanently stores user data and programs               | D. It connects the computer to external hardware devices          |

**Correct Answer: B - It directs and coordinates all CPU activities during execution**

**Why it is correct:** The Control Unit manages the fetch-decode-execute cycle and sends control signals so that the CPU, memory and other components perform the correct operations in the correct order.

**Why the other options are wrong:**

- A is mainly the job of the ALU.
- C is the role of secondary storage, not the Control Unit.
- D is handled by input/output interfaces and device controllers, not the primary function of the Control Unit.

**27. Based on the nature of its content, how can fake news be recognized?**

|                        |                           |
|------------------------|---------------------------|
| A. True information    | B. Misleading information |
| C. Educational content | D. Verified news          |

**Correct Answer: B - Misleading information**

**Why it is correct:** Fake news contains false, manipulated or misleading information presented as if it were genuine news.

**Why the other options are wrong:**

- A is not fake if it is accurate and properly presented.
- C can be genuine educational material.
- D has been checked or supported by credible verification, which is the opposite of fake news.

**28. What is the primary purpose of functions in JavaScript?**

|                                           |                                                                   |
|-------------------------------------------|-------------------------------------------------------------------|
| A. To store permanent data in the browser | B. To delete unwanted files from memory                           |
| C. To block advertisements on a webpage   | D. To group reusable blocks of code for performing specific tasks |

**Correct Answer: D - To group reusable blocks of code for performing specific tasks**

**Why it is correct:** A function packages instructions for a particular job. It can be called whenever needed, which reduces repetition and makes programs easier to organize, test and maintain.

**Why the other options are wrong:**

- A requires storage mechanisms such as localStorage or databases, not functions by themselves.
- B is not the normal purpose of JavaScript functions.



- C can be attempted by specialized code or extensions, but blocking advertisements is not the general purpose of functions.

**29. Why does a computer use secondary storage along with primary memory?**

|                             |                              |
|-----------------------------|------------------------------|
| A. To increase CPU speed    | B. To store data permanently |
| C. To control input devices | D. To process instructions   |

**Correct Answer: B - To store data permanently**

**Why it is correct:** Primary memory such as RAM is generally volatile, so its contents are lost when power is removed. Secondary storage such as an SSD or HDD keeps files and programs for long-term use.

**Why the other options are wrong:**

- A secondary storage is normally slower than RAM and does not directly increase CPU speed.
- C input devices are managed through controllers and system software.
- D instruction processing is performed by the CPU using primary memory.

**30. Which innovation has made it possible to conduct business and trade globally over the internet?**

|                      |                          |
|----------------------|--------------------------|
| A. Digital marketing | B. E-Commerce            |
| C. Social networking | D. Radio/TV broadcasting |

**Correct Answer: B - E-Commerce**

**Why it is correct:** E-commerce means buying and selling goods or services electronically over networks such as the internet, allowing businesses and customers in different countries to trade online.

**Why the other options are wrong:**

- A promotes products digitally but does not itself mean the online buying-and-selling transaction.
- C mainly supports social communication and networking.
- D is traditional broadcasting and does not provide the online commercial transaction model.

**31. Which ability helps entrepreneurs think of innovative and original business ideas?**

|                      |                       |
|----------------------|-----------------------|
| A. Coding ability    | B. Graphic designing  |
| C. Creative thinking | D. Marketing analysis |

**Correct Answer: C - Creative thinking**

**Why it is correct:** Creative thinking helps entrepreneurs generate new ideas, see problems from different perspectives and develop original products, services or business models.

**Why the other options are wrong:**

- A is a specific technical skill and is not required for every business idea.
- B is a visual-design skill, not the general innovation ability being asked.
- D is important for understanding markets but is not the same as generating original ideas.

**32. When a company collects and analyzes users private data without permission for profit, it leads to which issue?**

|                      |                          |
|----------------------|--------------------------|
| A. Data optimization | B. Privacy violation     |
| C. System efficiency | D. Digital collaboration |

**Correct Answer: B - Privacy violation**

**Why it is correct:** Collecting or using personal data without proper permission violates users privacy because information is being processed beyond what they knowingly agreed to.

**Why the other options are wrong:**

- A means improving how data is organized or used and does not describe the ethical problem.
- C concerns performance of a system.
- D means people or organizations working together using digital technology.

**33. A student spends most of the day on social media, resulting in poor academic performance and lack of sleep. Which harmful effect of computing does this reflect?**

|                          |                       |
|--------------------------|-----------------------|
| A. Enhanced productivity | B. Educational growth |
| C. Digital addiction     | D. Improved health    |

**Correct Answer: C - Digital addiction**

**Why it is correct:** Digital addiction is excessive or uncontrolled use of digital devices or services that interferes with normal life. Poor sleep and reduced study performance are typical negative consequences.

**Why the other options are wrong:**

- A and B would be positive outcomes, whereas the scenario shows harm.
- D is also a positive outcome and directly contradicts the lack of sleep described.

### 34. Why are registers regarded as the fastest type of memory in a computer system?

|                                                 |                                                                |
|-------------------------------------------------|----------------------------------------------------------------|
| A. They store large amounts of data permanently | B. They are built directly inside the CPU for immediate access |
| C. They are used for long-term data backup      | D. They are the least expensive type of memory                 |

**Correct Answer: B - They are built directly inside the CPU for immediate access**

**Why it is correct:** Registers are tiny storage locations inside the CPU. Because the processor can access them directly while executing instructions, they are faster than cache, RAM and secondary storage.

**Why the other options are wrong:**

- A is false because registers hold only very small amounts of temporary data.
- C is the purpose of secondary storage or backup media.
- D is false; very fast memory is generally expensive per unit of storage.

### 35. How does an algorithm help in solving a problem effectively?

|                                                   |                                                                   |
|---------------------------------------------------|-------------------------------------------------------------------|
| A. It organizes data in random order              | B. It converts programs into machine code automatically           |
| C. It deletes errors from a program automatically | D. It provides a clear step-by-step procedure to reach a solution |

**Correct Answer: D - It provides a clear step-by-step procedure to reach a solution**

**Why it is correct:** An algorithm describes a logical sequence of finite steps for solving a problem. It helps a student plan the solution before writing code.

**Why the other options are wrong:**

- A is the opposite of organized logical problem solving.
- B is the role of a compiler/interpreter or other language-processing software.
- C is not automatic; errors are found and corrected through debugging.

### 36. What is the primary purpose of JavaScript in web development?

|                                             |                                         |
|---------------------------------------------|-----------------------------------------|
| A. To structure the content of a webpage    | B. To style the appearance of a webpage |
| C. To make webpages interactive and dynamic | D. To design the layout of a webpage    |

**Correct Answer: C - To make webpages interactive and dynamic**

**Why it is correct:** JavaScript adds behavior to webpages, such as responding to button clicks, validating forms, changing content, performing calculations and displaying popups.

**Why the other options are wrong:**

- A is mainly the role of HTML.
- B is mainly the role of CSS.
- D is mainly handled through HTML structure and CSS layout rather than JavaScript as its primary purpose.

### 37. Which topology connects all devices to one central device?

|         |         |
|---------|---------|
| A. Bus  | B. Ring |
| C. Star | D. Mesh |

**Correct Answer: C - Star**

**Why it is correct:** In a star topology, each computer or network device has its own link to a central hub or switch.

**Why the other options are wrong:**

- A uses a shared backbone cable.
- B connects devices in a circular path.
- D provides multiple direct interconnections rather than relying on a single central device.

### 38. In computer networking, what are protocols defined as?

|                                      |                                                        |
|--------------------------------------|--------------------------------------------------------|
| A. Set of communication rules        | B. Hardware design specifications                      |
| C. Storage devices used in computers | D. Programming languages used for software development |

**Correct Answer: A - Set of communication rules**

**Why it is correct:** A network protocol is an agreed set of rules governing how devices format, send, receive and interpret data so that communication is possible.

**Why the other options are wrong:**

- B describes how hardware may be built, not how network messages are exchanged.
- C refers to physical storage hardware.
- D refers to languages used to write software, not the communication rules between network devices.

**39. Which technology enables everyday physical devices to connect to the internet and exchange data automatically?**

|                       |                      |
|-----------------------|----------------------|
| A. Internet of Things | B. Cloud Computing   |
| C. Embedded Systems   | D. Device Networking |

**Correct Answer: A - Internet of Things**

**Why it is correct:** The Internet of Things (IoT) connects physical objects such as sensors, appliances, vehicles and smart devices to networks so they can collect and exchange data automatically.

**Why the other options are wrong:**

- B provides remote computing and storage services but does not specifically mean connected everyday physical objects.
- C refers to specialized computers built into devices; embedded systems may work without internet connectivity.
- D is a general phrase rather than the recognized technology named in the curriculum.

**40. The 3 Vs of big data are:**

|                              |                                 |
|------------------------------|---------------------------------|
| A. Volume, Variety, Velocity | B. Value, Validity, Visibility  |
| C. Vision, Voice, Value      | D. Volume, Verification, Vision |

**Correct Answer: A - Volume, Variety, Velocity**

**Why it is correct:** The standard 3 Vs are Volume (large amount of data), Variety (many formats/types of data), and Velocity (the speed at which data is generated and processed).

**Why the other options are wrong:**

- B contains terms that may be discussed in data quality but are not the classic 3 Vs.
- C is not a standard big-data definition.
- D includes Volume but the other two terms are not the classic 3 Vs.

## Scenario-Based MCQs (41-60)

The model paper uses five scenarios: website development, JavaScript arrays, computer networks, data collection and a smart city. Read the scenario first, then connect the clue to the correct concept.

### 41. Scenario 1: Which software tool can edit website code?

|              |                  |
|--------------|------------------|
| A. Chrome    | B. Visual Studio |
| C. Cassandra | D. MongoDB       |

**Correct Answer: B - Visual Studio**

**Why it is correct:** Visual Studio is a development environment used to write, edit, run and debug code, including web-development code.

**Why the other options are wrong:**

- A is primarily a web browser.
- C is a database technology.
- D is a database management system, not a general website code editor.

### 42. Scenario 1: During system development, the system is organized into separate modules such as attendance, examinations and fees. This process is called:

|                |                  |
|----------------|------------------|
| A. Abstraction | B. Decomposition |
| C. Iteration   | D. Debugging     |

**Correct Answer: B - Decomposition**

**Why it is correct:** Decomposition breaks a large system or problem into smaller, manageable modules. Attendance, examinations and fees are separate subproblems of the larger school system.

**Why the other options are wrong:**

- A means focusing on important details and ignoring unnecessary ones.
- C means repeating a set of steps.
- D means finding and correcting errors.

### 43. Scenario 1: Voice-based controls such as saying "Attendance status" to open the attendance module are based on:

|                     |                             |
|---------------------|-----------------------------|
| A. Computer vision  | B. Speech Recognition (NLP) |
| C. Machine learning | D. Neural networks          |

**Correct Answer: B - Speech Recognition (NLP)**

**Why it is correct:** The system must recognize spoken words and interpret the users language command. Speech recognition combined with language processing is the direct technology described.

**Why the other options are wrong:**

- A interprets images and video.
- C can support speech systems internally, but it is a broader learning technique rather than the specific voice-control function.
- D may be used inside modern recognizers, but it is not the name of the user-facing speech-recognition technology.

### 44. Scenario 1: How does modular design support updating a specific part of a system like the exam section?

|                                                    |                                             |
|----------------------------------------------------|---------------------------------------------|
| A. By allowing changes in one module independently | B. It forces changes in the entire system   |
| C. By merging all modules into one program         | D. By converting code into machine language |

**Correct Answer: A - By allowing changes in one module independently**

**Why it is correct:** A major benefit of modular design is that one part can often be changed, tested or replaced without rewriting unrelated modules, which makes maintenance easier.

**Why the other options are wrong:**

- B is the opposite of modular design.
- C removes the separation that makes modules useful.
- D is compilation/translation and has nothing to do with modular updating.

### 45. Scenario 2: Given marks = [35, 50, 75, 100, 65], what will be printed by marks[1], marks[3] and marks.length?

|              |               |
|--------------|---------------|
| A. 35, 75, 5 | B. 50, 100, 5 |
|--------------|---------------|

C. 50, 75, 4

D. 35, 100, 4

**Correct Answer: B - 50, 100, 5**

**Why it is correct:** JavaScript arrays start indexing at 0. Therefore marks[1] is the second value 50, marks[3] is the fourth value 100, and marks.length is 5 because the array contains five elements.

**Why the other options are wrong:**

- A incorrectly treats marks[1] as 35 and marks[3] as 75.
- C uses the wrong fourth value and confuses length with the last index.
- D uses the wrong second value and also gives the last index rather than the number of elements.

**46. Scenario 2: Which JavaScript statement correctly adds the value 110 to the end of an array named marks?**

A. marks.add(110)

B. marks.push(110)

C. marks.insert(110)

D. marks.append(110)

**Correct Answer: B - marks.push(110)**

**Why it is correct:** The push() method adds one or more elements to the end of a JavaScript array.

**Why the other options are wrong:**

- A add() is not the standard JavaScript Array method for this operation.
- C insert() is not the standard JavaScript Array method here.
- D append() is not the standard JavaScript Array method for arrays.

**47. Scenario 2: What does marks.length return in a JavaScript array?**

A. Number of indexes starting from 0

B. Total number of elements in the array

C. Last index value of the array

D. Memory size of the array

**Correct Answer: B - Total number of elements in the array**

**Why it is correct:** The length property gives the number of elements. For five elements, length is 5 even though the last index is 4.

**Why the other options are wrong:**

- A is unclear and does not describe the property correctly.
- C confuses length with the last valid index, which is length - 1.
- D length does not report the memory size in bytes.

**48. Scenario 2: Which statement correctly calculates and displays the sum of the second and fourth elements of marks?**

A. document.write(marks[1] + marks[3]);

B. document.write(marks[2] + marks[4]);

C. document.write(marks.sum(1, 3));

D. document.write(marks.sum(2, 4));

**Correct Answer: A - document.write(marks[1] + marks[3]);**

**Why it is correct:** Because JavaScript indexing starts at 0, the second element is index 1 and the fourth is index 3. With the given data, the expression produces 50 + 100 = 150.

**Why the other options are wrong:**

- B adds the third and fifth elements instead.
- C uses a non-standard sum method and is not valid basic JavaScript array syntax.
- D has the same invalid sum-method issue and also uses the wrong positions.

**49. Scenario 3: The technician wants to reduce cable cost but accepts that a single cable failure may affect the whole network. Which topology fits this requirement?**

A. Bus topology

B. Ring topology

C. Star topology

D. Bus & Ring topologies both

**Correct Answer: A - Bus topology**

**Why it is correct:** A bus topology uses a shared main backbone cable and generally requires less cabling than a star. If that main backbone fails, communication across the network can be disrupted, matching the key clue in the question.

**Why the other options are wrong:**

- B forms a circular path; although a break can also disrupt a traditional ring, the description of low cable cost with a shared main cable most directly matches bus topology.
- C uses separate cables from each node to a central device, so it usually needs more cable.
- D is too broad because the question points to the single shared cable characteristic of bus topology.

**50. Scenario 3: Why is the central-device-based topology preferred in the school lab?**

|                       |                                               |
|-----------------------|-----------------------------------------------|
| A. It uses less cable | B. It does not require any device             |
| C. It is the cheapest | D. Failure of one node does not affect others |

**Correct Answer: D - Failure of one node does not affect others**

**Why it is correct:** In a star topology, every computer connects independently to the central hub or switch. If one computer or its cable fails, the other computers can normally continue working.

**Why the other options are wrong:**

- A is generally false compared with bus because star requires a separate link for each device.
- B is false because star specifically requires a central hub or switch.
- C is not always true and is not the reliability advantage emphasized by the scenario.

**51. Scenario 3: A student claims that if one computer fails in the central-device setup, the entire network will stop. What is the correct evaluation?**

|                                                 |                                                    |
|-------------------------------------------------|----------------------------------------------------|
| A. The statement is correct for star topology   | B. The statement is correct for ring topology      |
| C. The statement is incorrect for star topology | D. The statement is only correct for mesh topology |

**Correct Answer: C - The statement is incorrect for star topology**

**Why it is correct:** In a star network, each node has a separate connection to the central device, so failure of one computer does not normally stop the others. A failure of the central hub/switch is the critical failure point.

**Why the other options are wrong:**

- A is incorrect for the reason above.
- B describes a property of traditional ring networks, but the question asks about the central-device setup.
- D is incorrect because mesh provides multiple redundant paths and is not the setup described.

**52. Scenario 3: What is a key difference between star and ring topology in a school network?**

|                                                          |                                        |
|----------------------------------------------------------|----------------------------------------|
| A. Star topology has no connections between computers    | B. Ring topology uses a central device |
| C. Star topology uses a central device for communication | D. Both have identical structures      |

**Correct Answer: C - Star topology uses a central device for communication**

**Why it is correct:** The defining structure of a star topology is a central hub or switch. In a ring, devices are connected in a closed circular path rather than all depending on a central device.

**Why the other options are wrong:**

- A is misleading: star devices are connected through the central device.
- B incorrectly describes star topology.
- D is false because the physical/logical structures are different.

**53. Scenario 4: When data is collected from users through Google Forms or online questionnaires, it is called:**

|                                   |                        |
|-----------------------------------|------------------------|
| A. Sensor-based collection method | B. Survey method       |
| C. Simulation method              | D. Experimental method |

**Correct Answer: B - Survey method**

**Why it is correct:** Google Forms and online questionnaires ask respondents questions and record their responses, which is the survey method of data collection.

**Why the other options are wrong:**

- A uses physical or digital sensors to collect measurements automatically.
- C generates data by modeling a system rather than asking people questions.
- D gathers data through controlled experiments.

**54. Scenario 4: Why does the data science team remove duplicate and incomplete entries?**

|                              |                                         |
|------------------------------|-----------------------------------------|
| A. To increase storage usage | B. To improve data accuracy and quality |
| C. To reduce internet speed  | D. To randomize the dataset structure   |

**Correct Answer: B - To improve data accuracy and quality**

**Why it is correct:** Removing duplicate, missing or incomplete records is part of data cleaning. It reduces misleading counts and unreliable calculations, so later analysis becomes more accurate.

**Why the other options are wrong:**

- A is not a data-quality goal.
- C is unrelated to the contents of the dataset.

- D would make the dataset less orderly and does not improve quality.

**55. Scenario 4: Which data collection method automatically collects information in real time without human involvement?**

|            |                   |
|------------|-------------------|
| A. Surveys | B. Interviews     |
| C. Sensors | D. Questionnaires |

**Correct Answer: C - Sensors**

**Why it is correct:** Sensors can continuously detect and record events or measurements, such as attendance, temperature, movement or traffic, without a person manually entering each value.

**Why the other options are wrong:**

- A, B and D depend on human participants answering or providing information.

**56. Scenario 4: Which of the following is a primary data source in the scenario?**

|                     |                      |
|---------------------|----------------------|
| A. School databases | B. Published reports |
| C. Online surveys   | D. Historical books  |

**Correct Answer: C - Online surveys**

**Why it is correct:** Primary data is collected directly for the current study. Responses obtained directly from students or users through a survey are therefore primary data in this scenario.

**Why the other options are wrong:**

- A is an existing organizational record rather than newly collected survey responses.
- B is previously published material and is a secondary source.
- D is also previously existing information and is a secondary source.

**57. Scenario 5: What is the main reason smart cities rely on cloud computing for data storage?**

|                                              |                                                             |
|----------------------------------------------|-------------------------------------------------------------|
| A. To limit data sharing between departments | B. To enable flexible storage and easy access from anywhere |
| C. To reduce system functionality            | D. To avoid storing large amounts of data                   |

**Correct Answer: B - To enable flexible storage and easy access from anywhere**

**Why it is correct:** Cloud systems provide scalable storage and remote access. A smart city can expand capacity as data grows and allow authorized departments or services to access information from different locations.

**Why the other options are wrong:**

- A is opposite to the collaboration advantages of cloud services.
- C is not a goal of cloud computing.
- D is false because cloud computing is specifically useful for handling large amounts of data.

**58. Scenario 5: Why are cybersecurity measures important in a smart city system?**

|                                    |                                                           |
|------------------------------------|-----------------------------------------------------------|
| A. To increase traffic congestion  | B. To limit the use of technology                         |
| C. To reduce internet connectivity | D. To ensure data privacy and prevent unauthorized access |

**Correct Answer: D - To ensure data privacy and prevent unauthorized access**

**Why it is correct:** Smart-city systems collect and transmit sensitive information. Cybersecurity protects systems and data from unauthorized access, theft, manipulation and disruption.

**Why the other options are wrong:**

- A, B and C reduce usefulness or performance and are not security objectives.

**59. Scenario 5: What is the main purpose of installing sensors on roads in a smart city?**

|                                         |                                      |
|-----------------------------------------|--------------------------------------|
| A. To store citizen personal data       | B. To collect real-time traffic data |
| C. To replace traffic police completely | D. To provide internet to drivers    |

**Correct Answer: B - To collect real-time traffic data**

**Why it is correct:** Road sensors measure traffic flow, density, speed or congestion as it happens. This real-time data can then be analyzed to control signals and recommend routes.

**Why the other options are wrong:**

- A is not the stated traffic-management purpose.
- C is too absolute and not the function of the sensor itself.
- D is a connectivity service, not the purpose of a traffic sensor.



**60. Scenario 5: A driver receives AI-based suggestions for alternate routes during traffic congestion. What kind of action is the system performing?**

|                                         |                                                        |
|-----------------------------------------|--------------------------------------------------------|
| A. To permanently store traffic records | B. Predicting and optimizing traffic flow in real time |
| C. To delete GPS data from vehicles     | D. To increase fuel consumption                        |

**Correct Answer: B - Predicting and optimizing traffic flow in real time**

**Why it is correct:** The AI analyzes current traffic data, predicts conditions and recommends a better route. This is real-time prediction and optimization aimed at reducing congestion and travel time.

**Why the other options are wrong:**

- A is storage, not the decision-making action described.
- C would remove useful navigation data rather than optimize routes.
- D is the opposite of the intended benefit.



## SECTION A - FINAL ANSWER KEY

| Q  | Ans | Q  | Ans | Q  | Ans |
|----|-----|----|-----|----|-----|
| 1  | C   | 21 | A   | 41 | B   |
| 2  | A   | 22 | A   | 42 | B   |
| 3  | B   | 23 | A   | 43 | B   |
| 4  | C   | 24 | C   | 44 | A   |
| 5  | C   | 25 | B   | 45 | B   |
| 6  | B   | 26 | B   | 46 | B   |
| 7  | C   | 27 | B   | 47 | B   |
| 8  | B   | 28 | D   | 48 | A   |
| 9  | D   | 29 | B   | 49 | A   |
| 10 | C   | 30 | B   | 50 | D   |
| 11 | B   | 31 | C   | 51 | C   |
| 12 | C   | 32 | B   | 52 | C   |
| 13 | B   | 33 | C   | 53 | B   |
| 14 | C   | 34 | B   | 54 | B   |
| 15 | A   | 35 | D   | 55 | C   |
| 16 | B   | 36 | C   | 56 | C   |
| 17 | B   | 37 | C   | 57 | B   |
| 18 | C   | 38 | A   | 58 | D   |
| 19 | D   | 39 | A   | 59 | B   |
| 20 | C   | 40 | A   | 60 | B   |

## SECTION B - COMPLETE SOLUTIONS (INCLUDING OR)

Section B contains six 3-mark questions. A strong 3-mark response should be direct, accurate and organized in points, with a suitable example where required.

### Question 1: What is a business plan? Write down its two components.

A business plan is a written document that explains a business idea, its objectives, how the business will operate, how it will attract customers, and how it expects to earn money. In simple words, it is a roadmap showing how a business will be started, managed and made successful.

Two important components are:

- Marketing Plan - identifies target customers, competitors, pricing, promotion and how the product or service will reach the market.
- Financial Plan - estimates startup cost, investment, income, expenses, cash needs and expected profit.

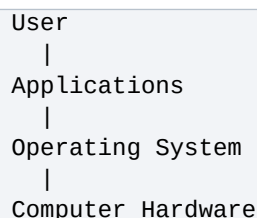
**Exam-ready answer:** A business plan is a written document that describes the goals of a business and explains how those goals will be achieved. Two important components are the marketing plan, which describes customers and promotion strategies, and the financial plan, which describes costs, investment, income and expected profit.

### OR: What is an operating system? Give any two examples.

An operating system (OS) is system software that manages computer hardware and software resources and provides an interface between the user, applications and the computer hardware. It manages functions such as processor use, memory, files and input/output devices.

- Example 1: Microsoft Windows
- Example 2: Linux

Other valid examples include macOS, Android and iOS.



### Question 2: Compare Simplex and Full Duplex modes of communication with one example each.

| Feature                    | Simplex                  | Full Duplex                       |
|----------------------------|--------------------------|-----------------------------------|
| Direction                  | One direction only       | Both directions                   |
| Simultaneous communication | No                       | Yes                               |
| Sender/receiver roles      | Fixed                    | Both devices can send and receive |
| Example                    | TV or radio broadcasting | Telephone call                    |

Simplex example: A television station sends information to a TV set; the TV receives but does not send the same communication back to the station.

Full Duplex example: During a telephone call, both people can speak and listen at the same time.

### OR: Differentiate between discrete and continuous data with one example of each.

| Feature      | Discrete Data                     | Continuous Data                        |
|--------------|-----------------------------------|----------------------------------------|
| Nature       | Countable, separate values        | Measurable values over a range         |
| How obtained | Usually by counting               | Usually by measuring                   |
| Typical form | Often whole-number counts         | Can include fractions/decimals         |
| Example      | Number of students: 30, 31, 32... | Height: 170 cm, 170.2 cm, 170.25 cm... |

A class cannot normally contain 30.5 students, so the count of students is discrete. Height can take many values within a range, so it is continuous.

### Question 3: How can a user search for information on the internet in a safe and responsible way? State any three methods.

1. Use reliable and trusted websites such as educational institutions, government departments and reputable organizations. Verify important information from more than one credible source.
2. Protect personal information. Do not enter passwords, banking details, home address or other sensitive information on suspicious websites or through unknown links.
3. Check information before using or sharing it. Look at the author, date, source, supporting evidence and whether the claim can be confirmed elsewhere. Do not spread fake or misleading information.

**Exam-ready answer:** A user can search safely and responsibly by using reliable websites, protecting personal information from suspicious sites and links, and verifying information from multiple authentic sources before accepting or sharing it.

### OR: Compare simple and complex problems with one daily-life example of each.

| Feature            | Simple Problem                         | Complex Problem                              |
|--------------------|----------------------------------------|----------------------------------------------|
| Steps              | Few                                    | Many                                         |
| Understanding      | Easy and direct                        | More difficult, with many factors            |
| Strategy           | Often solved directly                  | Often needs decomposition into smaller tasks |
| Daily-life example | Finding the total price of three books | Organizing a school annual function          |

A school annual function is complex because it involves budget, venue, invitations, schedule, equipment, students, teachers and security. Dividing it into smaller responsibilities makes it manageable.

### Question 4: How does design thinking help a startup create innovative solutions? Give any three points.

4. It helps the team understand real user needs. By talking to users and observing their problems, the team avoids building features that users do not need.
5. It encourages creative ideas. The team generates several possible solutions instead of accepting the first idea, which increases the chance of innovation.
6. It uses prototypes and testing. A simple version is tested with users, feedback is collected, and the design is improved before the final product is developed.

**Exam-ready answer:** Design thinking creates innovative solutions because it focuses on actual user needs, encourages different creative ideas, and allows prototypes to be tested and improved according to user feedback before the final product is developed.

### OR: A teacher uses graphs and charts to analyze student exam results. Explain any two applications of data visualization in this situation.

7. Comparing performance - a bar chart can compare marks of different students or subjects, making high and low scores visible at a glance.
8. Identifying trends and making decisions - graphs can show improvement, decline, difficult subjects or weak students. The teacher can use these patterns to arrange extra support or change teaching strategies.

**Exam-ready answer:** Data visualization can be used to compare marks of students or subjects using graphs and charts and to identify performance trends, weak areas and high-performing students so that appropriate teaching decisions can be made.

### Question 5: Write down the purpose of any three input devices.

| Input Device | Purpose                                                                                |
|--------------|----------------------------------------------------------------------------------------|
| Keyboard     | Enters letters, numbers, symbols and commands into the computer.                       |
| Mouse        | Controls the pointer and allows the user to point, click, select and drag objects.     |
| Microphone   | Enters voice and other sound into the computer by converting sound into digital input. |

**OR: List any three applications of Artificial Intelligence in real life.**

9. Voice assistants - systems such as Siri, Google Assistant or Alexa understand voice commands and provide responses.
10. Face recognition - smartphones and security systems identify or verify users by analyzing facial features.
11. Recommendation systems - websites and applications suggest videos, movies or products based on a users previous activity and preferences.

Other valid examples include medical diagnosis, smart traffic systems, fraud detection, chatbots and self-driving technologies.

**Question 6: Identify and correct errors in the JavaScript code.**

Code given in the paper:

```
let fruits = ["Apple", "Banana", "Orange"];
console.log(fruits.length)
console.log(fruit.length);
```

12. Missing quotation mark: "Banana must have a closing quotation mark before the comma. Correct form: "Banana",.
13. Wrong variable name: the array is named fruits, but the last statement uses fruit. JavaScript treats fruit and fruits as different identifiers.
14. For clean exam syntax, write a semicolon after console.log(fruits.length); as well. JavaScript can sometimes insert semicolons automatically, so the first two errors are the critical ones.

Fully corrected code:

```
let fruits = ["Apple", "Banana", "Orange"];
console.log(fruits.length);
console.log(fruits.length);
```

Output: 3 on the first line and 3 on the second line because the array contains three elements.

**OR: Compare any two types of storage devices with respect to reliability, capacity and speed.**

A strong comparison is HDD versus SSD:

| Feature                                                                                                                                                                                                                                                   | HDD                                                    | SSD                                                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|---------------------------------------------------------------------|
| Technology                                                                                                                                                                                                                                                | Magnetic rotating disks and moving parts               | Flash memory with no moving parts                                   |
| Reliability                                                                                                                                                                                                                                               | Lower resistance to shock because of moving parts      | Generally more resistant to shock because there are no moving parts |
| Capacity                                                                                                                                                                                                                                                  | Very large capacities available at relatively low cost | Large capacities available but often more expensive per GB          |
| Speed                                                                                                                                                                                                                                                     | Slower                                                 | Much faster                                                         |
| <b>Exam-ready comparison:</b> An HDD offers high capacity at low cost but is slower and mechanically less robust. An SSD has no moving parts, is much faster and generally more resistant to shock, although high-capacity SSDs usually cost more per GB. |                                                        |                                                                     |

## SECTION C - COMPLETE SOLUTIONS (INCLUDING OR)

Section C contains two 6-mark questions. Full-mark answers should contain a clear introduction, several well-explained points, relevant examples and a brief conclusion where appropriate.

### Question 7: Explain how computing innovations can negatively affect society and the environment. Support your answer with real-life examples.

Computing innovations have greatly improved communication, education, business and daily life. However, irresponsible design or excessive use can also create social and environmental problems.

- 1. Privacy violations:** Websites and applications may collect location, browsing activity, contacts or other personal data without proper consent. Example: an app tracks a user's behavior and shares it for advertising without clearly informing the user.
- 2. Cybercrime:** Technology can be misused for hacking, phishing, online fraud, identity theft and malware attacks. Example: a fake banking email tricks a user into entering a password on a fraudulent website.
- 3. Digital addiction:** Excessive use of social media, games or smartphones can reduce study time, disturb sleep and affect concentration. Example: a student spends many hours online each day and performs poorly in class.
- 4. Spread of fake information:** False or manipulated information can spread quickly through social media. Example: an unverified story is shared thousands of times and creates confusion before it is corrected.
- 5. Electronic waste:** Old phones, computers, batteries and monitors become e-waste. Improper disposal can release harmful materials into soil and water. Example: batteries dumped in an open landfill may leak hazardous substances.
- 6. High energy consumption:** Data centers and large computing systems consume significant electricity for both computing and cooling. If the electricity comes from fossil fuels, this contributes indirectly to carbon emissions.

**Conclusion:** Computing can negatively affect society through privacy violations, cybercrime, digital addiction and misinformation, while environmental effects include e-waste and high energy use. Responsible use, cybersecurity, ethical data practices and proper recycling can reduce these harms.

### OR: Explain how the OSI model helps manage communication in a school network by describing the role of any three layers.

OSI stands for Open Systems Interconnection. It is a seven-layer reference model that explains how data moves from one network device or application to another. Dividing communication into layers makes networking easier to design, standardize and troubleshoot.

7. Application
6. Presentation
5. Session
4. Transport
3. Network
2. Data Link
1. Physical

Any three layers may be described. The following three are detailed and easy to relate to the school scenario:

- 1. Application Layer:** This layer is closest to the user and provides network services to applications, such as web access, email or file sharing. School example: a teacher in the library uses a file-sharing application to access a document on another computer.
- 2. Transport Layer:** This layer manages end-to-end delivery. It can divide information into smaller segments, help control delivery and ensure that data is reassembled correctly. School example: when a large file is transferred from the administrative office to the lab, the transport layer helps the complete data reach the destination reliably.
- 3. Network Layer:** This layer handles logical addressing and routing so packets can reach the correct destination. School example: data sent from a lab computer to an administrative-office computer is directed toward the correct network address.

**Why the OSI model helps:** Each layer has a defined responsibility. If communication fails, the network administrator can check whether the problem is physical connectivity, addressing/routing, reliable delivery or the user application instead of treating networking as one confusing process.

## Question 8: Create a result sheet using HTML that displays the given data in table format.

The paper gives five student records with Math, English, Computer Science, Total Marks and Percentage. A complete HTML answer is:

```
<!DOCTYPE html>
<html>
<head>
  <title>Student Result Sheet</title>
</head>
<body>

<h2>Student Result Sheet</h2>

<table border="1">
  <tr>
    <th>Roll No</th>
    <th>Student Name</th>
    <th>Math</th>
    <th>English</th>
    <th>Computer Science</th>
    <th>Total Marks</th>
    <th>Percentage</th>
  </tr>
  <tr>
    <td>1</td><td>Ali Ahmed</td><td>78</td><td>85</td>
    <td>90</td><td>253</td><td>84.3%</td>
  </tr>
  <tr>
    <td>2</td><td>Sara Khan</td><td>88</td><td>79</td>
    <td>92</td><td>259</td><td>86.3%</td>
  </tr>
  <tr>
    <td>3</td><td>Usman Ali</td><td>65</td><td>72</td>
    <td>70</td><td>207</td><td>69.0%</td>
  </tr>
  <tr>
    <td>4</td><td>Ayesha Noor</td><td>90</td><td>88</td>
    <td>95</td><td>273</td><td>91.0%</td>
  </tr>
  <tr>
    <td>5</td><td>Hamza Raza</td><td>55</td><td>60</td>
    <td>68</td><td>183</td><td>61.0%</td>
  </tr>
</table>

</body>
</html>
```

Important tags to understand:

| Tag        | Purpose                        |
|------------|--------------------------------|
| <table>    | Starts a table                 |
| <tr>       | Creates a table row            |
| <th>       | Creates a heading cell         |
| <td>       | Creates a normal data cell     |
| border="1" | Displays a simple table border |

The table produced by the code contains the following data:

| Roll No | Student Name | Math | English | Computer Science | Total Marks | Percentage |
|---------|--------------|------|---------|------------------|-------------|------------|
| 1       | Ali Ahmed    | 78   | 85      | 90               | 253         | 84.3%      |
| 2       | Sara Khan    | 88   | 79      | 92               | 259         | 86.3%      |
| 3       | Usman Ali    | 65   | 72      | 70               | 207         | 69.0%      |
| 4       | Ayesha Noor  | 90   | 88      | 95               | 273         | 91.0%      |
| 5       | Hamza Raza   | 55   | 60      | 68               | 183         | 61.0%      |

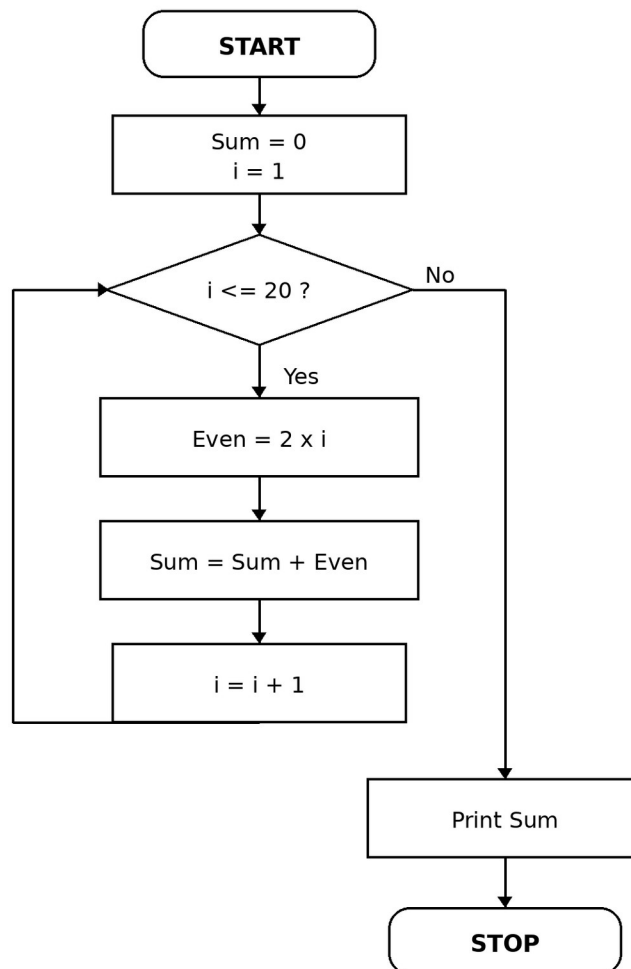
**OR: Draw a flowchart that prints the sum of the first 20 even numbers.**

The first 20 even numbers are 2, 4, 6, ..., 40. A convenient way to generate them is  $\text{Even} = 2 \times i$ , where  $i$  runs from 1 to 20.

Algorithm:

1. START
2. Set  $\text{Sum} = 0$  and  $i = 1$
3. Check: Is  $i \leq 20$ ?
4. If Yes, set  $\text{Even} = 2 \times i$
5. Set  $\text{Sum} = \text{Sum} + \text{Even}$
6. Set  $i = i + 1$  and return to Step 3
7. If No, print Sum
8. STOP

Flowchart:



Verification of the final answer:

$$\begin{aligned}
 &2 + 4 + 6 + \dots + 40 \\
 &= 2(1 + 2 + 3 + \dots + 20) \\
 &= 2 \times [20 \times 21 / 2] \\
 &= 420
 \end{aligned}$$

**Final output:** The flowchart prints 420.



## EXAM PREPARATION CHECKLIST

This model paper repeatedly tests the following concepts. Students should be able to define them, recognize them inside scenarios and apply them in short problems:

- Computational Thinking - algorithm, decomposition, abstraction, pattern recognition and debugging.
- Web Development - HTML structure, CSS styling and JavaScript interactivity.
- JavaScript - arrays, zero-based indexing, length, push(), functions and alert().
- Artificial Intelligence - NLP, speech recognition, computer vision, digital assistants and smart-city applications.
- Data Science - qualitative data, discrete/continuous data, primary data, surveys, sensors, data cleaning, visualization and big data.
- Networking - simplex/full duplex, protocols, bus/star/ring topologies, IoT and the OSI model.
- Computer Systems - RAM, registers, Control Unit, secondary storage, operating systems and input devices.
- Digital Citizenship - phishing, fake news, software piracy, privacy, digital addiction, cybersecurity and responsible internet use.

**Final advice:** Do not memorize answer letters only. Read the clue in each scenario and ask: Which concept is being demonstrated? This is the most reliable way to handle the new FBISE scenario-based pattern.

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*For educational use and student exam preparation.*